

Resolutions of a league of 4 with football points

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September 2026

1 Introduction

Most international football competitions of national teams, like the World Cup or the European Championship, start with a league stage in which each league¹ has 4 participants.

2 League rules

The leagues follow a round-robin format, i.e. every team plays any other team exactly once. Note that we are assuming that the games are played in a neutral venue, as is the case in national teams competitions. In club competitions, it is usual to play the games in the clubs' own stadiums, and then clubs face each other twice: one home and one away. We will carry out the analysis for the national teams case, and then generalise to the clubs format.

Like in any official football competition, teams are awarded 3 points per win, 1 point per draw and 0 points per loss. The standings are then ordered by the total number of points obtained. Tie-break rules based on goals are defined for teams having scored the same amount of points. However, here we will only look at the point distribution, hence we do not need to break ties.

Formalism

Let $n > 1$ be an integer, and consider the league of n teams. The teams will be denoted A_i for $i \in \{1, \dots, n\}$. The result of the game between team A_i

¹We use the term *league* instead of the equally common *group* to avoid confusion with mathematical groups.

and team A_j will be indicated by an element $a_{ij} \in \mathbb{F}_3 = \{-1, 0, 1\}$. Namely, we set

$$a_{ij} = \begin{cases} 1 & \text{if } A_i \text{ wins,} \\ 0 & \text{if the game is a draw,} \\ -1 & \text{if } A_j \text{ wins.} \end{cases}$$

Notice that, for $i \neq j$, both a_{ij} and a_{ji} are well-defined and satisfy $a_{ij} = -a_{ji}$. Given a result a_{ij} , we will say that A_i is the *main team* of that presentation of the game.

Furthermore, let us set $a_{ii} = 0$ for all i . This has no meaning as a team does not play against itself. Then, we can encode all results in an antisymmetric matrix $A = (a_{ij})$, which we call the *league matrix*. Note that the set of antisymmetric matrices of size n forms a group $\mathbf{Sk}(n)$ with respect to the sum of matrices. Indeed, the set is closed under the sum, contains the zero matrix and all inverses. Even more, $\mathbf{Sk}(n)$ is a vector space, with dimension equal to the number of games,

$$\dim \mathbf{Sk}(n) = \binom{n}{2} = \frac{n(n-1)}{2}.$$

Thus, the total number of possible league matrices for a league with n teams equals

$$N(n) := 3^{\frac{n(n-1)}{2}}. \quad (1)$$

Given a league matrix A , we may obtain the league standings vector $P(A)$, of dimension n , by computing the number of points p_i for each team A_i and then sorting the entries in descending order. For that, let us first define the function $p : \mathbb{F}_3 \rightarrow \mathbb{N}$ that yields the points scored by the main team of a given game result a . We want p to map $1 \mapsto 3$, $0 \mapsto 1$ and $-1 \mapsto 0$, and we get that with the polynomial expression

$$p(a) = 1 + \frac{3}{2}a + \frac{1}{2}a^2.$$

With that, we set

$$p_i = \sum_{j \neq i} p(a_{ij}),$$

and $P(A)$ as the vector containing the p_i in decreasing order.

3 League resolutions

In this work, we are interested in classifying the possible standings types of a league. Specifically, we seek to establish two different classifications. First, we wish to compute all different standings vectors $P(A)$ for a given amount of teams. In other words, we are looking for the cardinality of the quotient set

$$\mathbf{Sk}(n)/\sim_{\text{std}}, \quad (2)$$

where the equivalence relation $\sim_{\text{std}}$ is defined by

$$A \sim_{\text{std}} B \iff P(A) = P(B), \quad (3)$$

for $A, B \in \mathbf{Sk}(n)$. We will denote this cardinal by

$$\text{Std}(n).$$

Second, we aim to obtain a finer classification, which identifies league matrices only when their sequences of results are *isomorphic*. By this, we understand that the second league can be obtained from the first by applying some permutation to the set of teams. In other words, the structure of results was exactly the same in both leagues – they were just labelled differently. To capture that, we consider the natural action $S_n \curvearrowright \mathbf{Sk}(n)$ given by matrix multiplication. Here, each permutation $\sigma \in S_n$ is represented by the $n \times n$ matrix (m_{ij}) , where

$$m_{ij} = \begin{cases} 1 & \text{if } i = \sigma(j), \\ 0 & \text{otherwise.} \end{cases}$$

The orbits of this action define a partition of $\mathbf{Sk}(n)$, and we write the quotient set as

$$\mathbf{Sk}(n)/S_n.$$

Hence, two matrices $A, B \in \mathbf{Sk}(n)$ are in the same *isomorphism* class, $A \sim_{\text{iso}} B$, if there exists a permutation $\sigma \in S_n$ such that

$$\sigma \cdot A = B.$$

The cardinal of the quotient set $\mathbf{Sk}(n)/S_n$ will be denoted by

$$\text{Iso}(n).$$

It follows that $\sim_{\text{iso}}$ is a refinement of $\sim_{\text{std}}$. Indeed, any two isomorphic leagues give rise to the same standings vector. The contrary, however, is not true in general, although it holds for $n \in \{2, 3\}$. We show these results below.

3.1 The case of one game ($n = 2$)

When $n = 2$, the league reduces to a single game between teams A_1 and A_2 , with three possible outcomes, i.e. $N(2) = 3$ as we can check from Eq. (1). In other words, the dimension of the vector space $\mathbf{Sk}(2)$ equals 1. It is clear that $\mathbf{Std}(2) = 2$, since the point distribution is different whether there is a draw (11) or one of the teams wins the game (30). In turn, the two cases without draw are equivalent after applying the permutation (12), and we conclude that $\mathbf{Iso}(2) = 2$ as well.

3.2 The case $n = 3$

If $n = 3$, we have 3 teams and 3 games are played. The combinatorics become now notably harder.

3.3 The case $n = 4$

3.4 Higher cases